

SUPPLY CHAIN & OPERATIONS ANALYTICS

DataCo shipping and fulfilment performance

Portfolio Project - reviewed descriptive analysis with documented source limitations.

Coverage: 62,897 eligible distinct orders from Jan 2015 to Jan 2018; 36,048 labeled late (57.3%). 2,855 canceled orders are reported separately.

Source: DataCo Smart Supply Chain for Big Data Analysis, Mendeley Data, version 5 (12 March 2019), CC BY 4.0. <https://data.mendeley.com/datasets/8gx2fvg2k6/5>

Three leading findings

1. Geographic rates are closely grouped: market rates span only 56.5%-57.7%; Europe contributes the largest late-order count (10,199). Pacific Asia has the highest market rate (57.7%; 9,720/16,853). Central Africa is the highest-rate region (60.0%; 533 eligible), while Canada is lowest (52.6%).
2. Mode results differ, but are not causal: First Class is highest by labeled late rate (100.0%; 9,602/9,602); Standard Class is lowest (39.8%). All 9,602 eligible First Class orders record `actual_days` = 2 and `scheduled_days` = 1. This uniform recorded pattern requires definition/source validation; no cause is established. Mode commitments and order mixes differ.
3. No sustained improvement or deterioration is demonstrated: full-year rates are 57.2% in 2015, 57.6% in 2016, and 57.0% in 2017. The weighted three-month rate was 57.2% by Mar 2015 and 58.1% by Jan 2018. Changing market coverage limits these global comparisons; January 2018 contains only Pacific Asia. These pooled rates do not establish a stable-market trend.

Purpose: identify where late shipping is concentrated, compare observed performance against recorded service durations, and track reliability over order month. No causal claims are made.

1. Source, structure & quality

The supplied file contains 180,519 order-item rows and 53 columns. Each Order Item Id is unique; there are 65,752 orders, 20,652 customers, and 118 product IDs. The order-level dimensions and shipping fields checked were consistent across all item rows within each order. There are no exact duplicate rows.

Order dates cover 2015-01-01 to 2018-01-31; shipping timestamps extend to 2018-02-06. Date and delivery-status fields are complete in the parsed data. 36,787 orders have an Order Status other than COMPLETE/CLOSED; status names are not treated as a reliable binary completion flag for the primary population.

Key quality observations

Field	Missing	Missing %	Distinct
Order Zipcode	155,679	86.24%	609
Customer Zipcode	3	0.00%	995
Customer Lname	8	0.00%	1,109
Product Description	180,519	100.00%	0
Product Status	0	0.00%	1

Order Zipcode is 86.2% missing (exact missing rate shown in workbook); Product Description is empty throughout and Product Status is constant. Text whitespace was trimmed. Geographic labels remain in their original language; only surrounding whitespace was removed. Email/password values are masked in the source, but all names, emails, passwords, streets, customer IDs, customer coordinates, and other unnecessary personal fields are omitted from cleaned outputs. No postcodes or durations are imputed.

Financial checks: all 180,519 Sales values reproduce quantity times price within 0.01 source monetary units. Item Total differs from Sales minus discount by more than 0.01 in 1,224 rows; the maximum residual is 0.010014, consistent with boundary rounding and floating-point precision. Recorded discount-rate and profit-ratio identities have maximum residuals of about 10.00 and 9.30 units, so rounding alone is not a sufficient blanket explanation. Negative profit occurs in 33,784 item rows (18.7%), with similar shares across shipping outcomes; retain these observations. Currency and exact ratio definitions remain undocumented; no global financial comparison is made.

2. Analytical framework

Primary eligible population: one distinct order per Order Id with Delivery Status in Advance shipping, Shipping on time, or Late delivery. Late rate = eligible orders labeled Late delivery / eligible distinct orders. Shipping canceled is excluded from denominators and shown separately. The risk flag agrees exactly with Delivery Status in this file.

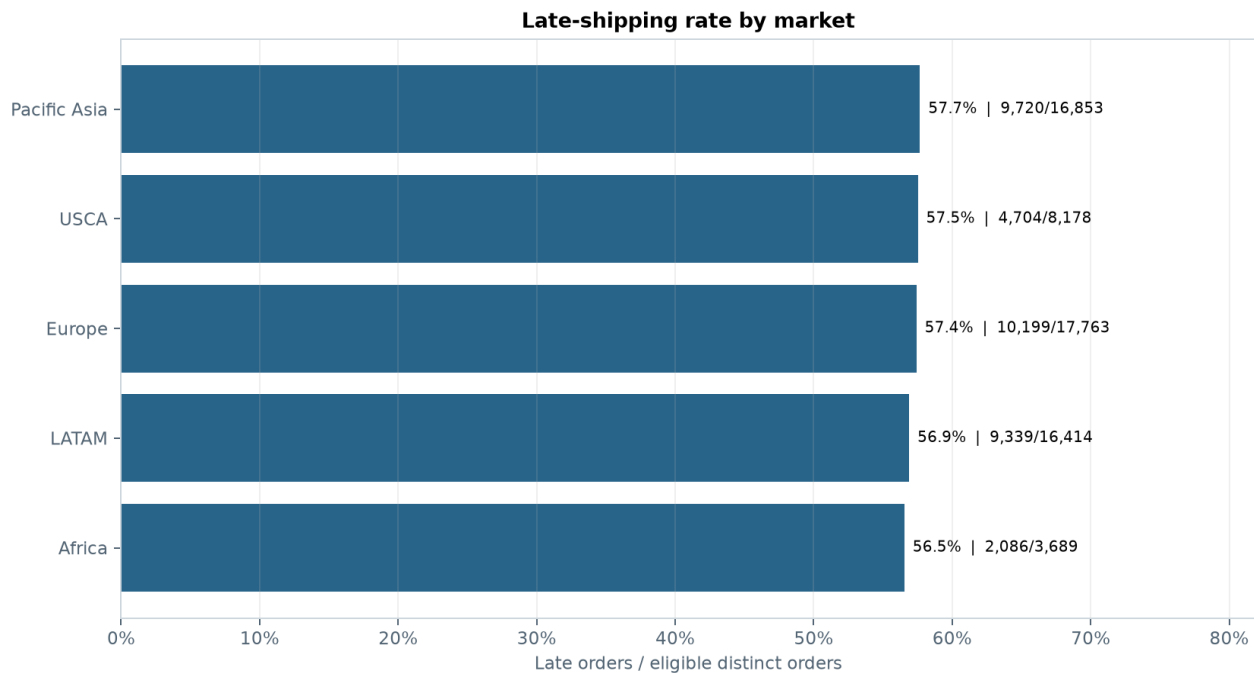
Uncertainty and volume: 95% Wilson intervals are shown for group and monthly rates. Groups below 100 eligible orders are flagged low volume. The threshold is a reporting guardrail, not a significance test. Geographic differences are also checked within recorded shipping modes and years.

Duration and time: recorded gap = actual shipping days – scheduled shipment days. Order month is the initial trend basis. A weighted three-month rate sums late orders divided by eligible orders across three months. Shipping timestamps are shipment timestamps, not proven delivery-arrival timestamps.

Robustness: COMPLETE/CLOSED among primary eligible orders is a sensitivity population only. Findings are compared on monthly/quarterly views, excluding final month, by mode mix, and in coverage-qualified fixed-market windows. Associations do not establish the effect of switching mode.

3. Geographic shipping performance

Overall primary late rate: $36,048/62,897 = 57.3\%$ (95% Wilson interval 56.9%-57.7%). Market and region tables distinguish rate from volume; group shares are shares of the 36,048 eligible late orders.



By region, Central Africa has the highest rate at 60.0% (320/533); its group is not low volume. The chart and CSV include eligible counts, late counts, confidence limits, share of late orders, and percentage-point difference from overall.

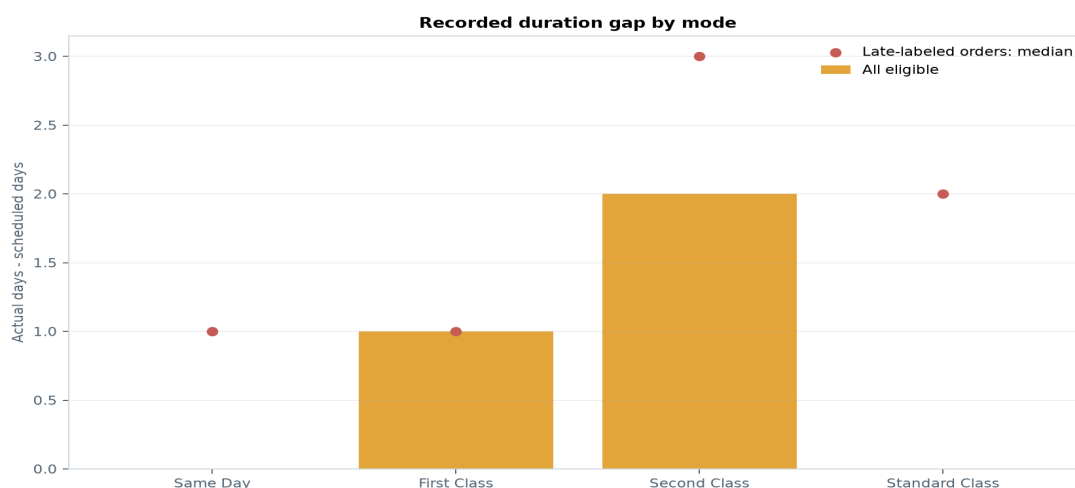
Europe has 10,199 late orders (28.3% of all late orders), followed by Pacific Asia with 9,720 (27.0%). Western Europe contributes 5,585 late orders, the largest region count. Market rates differ by only 1.13 percentage points; their Wilson intervals overlap. After applying the same overall mode weights to every market, rates span 57.0%-57.7% (0.62 points). Africa moves from 56.5% observed to 57.3% adjusted, illustrating mix sensitivity. Central Africa has the highest region rate, 60.0% (320/533; interval 55.8%-64.1%), but much less volume than Western Europe. Market-year coverage is uneven: Africa has no 2015 orders, LATAM has no 2016 orders, and January 2018 contains only Pacific Asia. No market is highest in every observed full year or in every mode; geography ranks should not be treated as a stable performance hierarchy. Standardization is descriptive and does not establish causal effects.

What this means in simple terms... The rate tells us how often a market or region is late, while the count tells us how many orders that represents. Both matter when prioritizing investigation.

4. Shipping-mode performance



First Class has the highest late-labeled rate (100.0%); Standard Class the lowest (39.8%). Observed overall median duration gaps range by mode; median gaps appear below; interquartile ranges and full distributions are in the workbook and CSVs. All 9,602 eligible First Class orders have `actual_days` = 2 and `scheduled_days` = 1. This uniform recorded pattern requires definition/source validation; it does not establish a cause. Recorded gaps are not arrival timestamp measures.



Among the 36,048 late-labeled orders, 36,048 have actual days greater than scheduled days and 0 have a zero or negative recorded gap. All late-labeled eligible orders have positive gaps. Among eligible orders, the late flag exactly equals actual days greater than scheduled days; canceled orders remain separate. Same Day records: 3,571/3,571 have a 12-hour order-to-shipment timestamp interval; recorded actual duration values are 0 day(s): 1,844, 1 day(s): 1,727. These records do not establish delivery within 12 hours: the timestamp is shipment time, and integer actual days and outcome labels follow separate conventions.

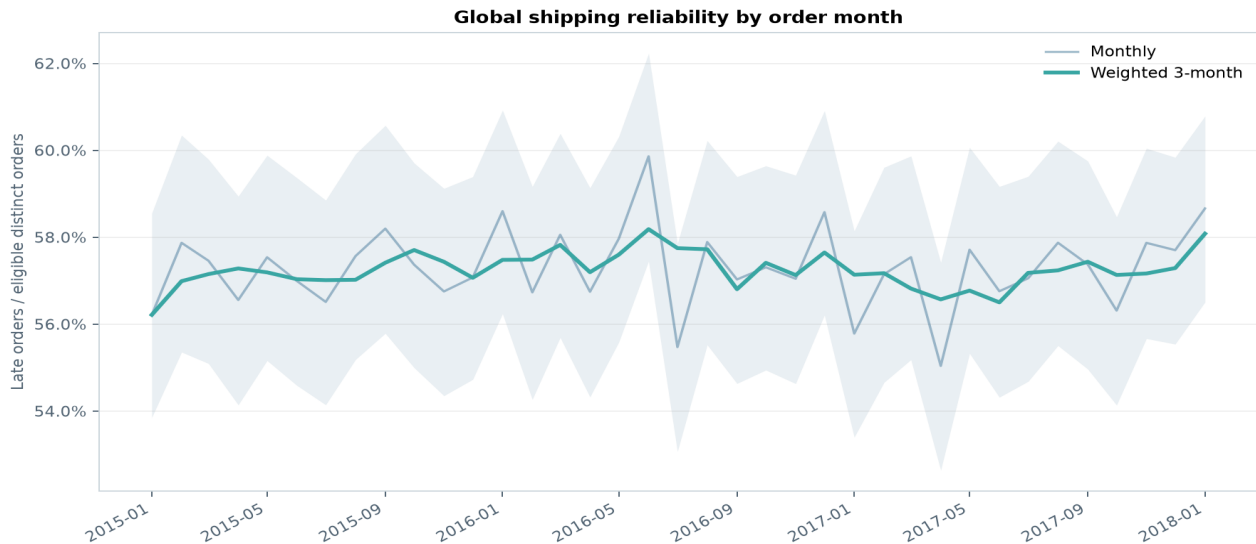
By mode, market, and year comparison tables are available in the workbook and CSVs. Mode assignment is observational: service commitments, destinations, and order mixes differ, so this analysis does not claim that switching modes would improve performance.

What this means in simple terms... A mode's observed late rate is not a fair promise comparison by itself. Pair the label with the scheduled-versus-actual duration and compare like markets and periods before changing service rules.

5. Reliability over time

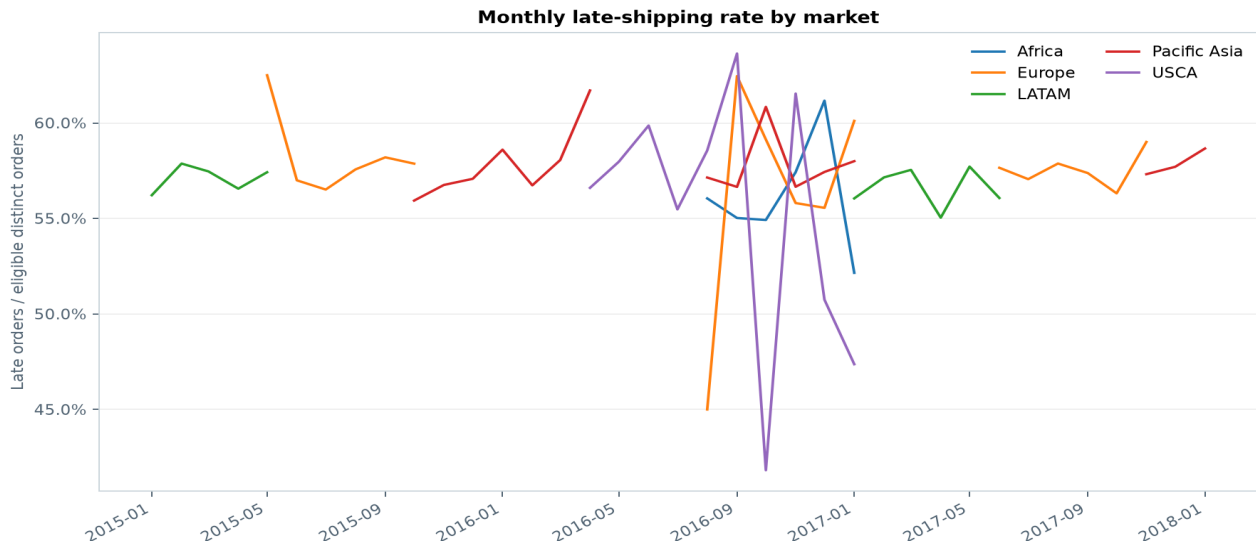
Geographic coverage changes over time, limiting global comparisons. January 2018 contains only Pacific Asia; absent market-months are not zero late rates. Pooled rates describe changing observed cohorts, not like-for-like global performance.

Changing market coverage limits global comparisons; January 2018 is Pacific Asia only.



Order-month late rate changed from 56.2% (963/1,713) in Jan 2015 to 58.7% (1,195/2,037) in Jan 2018. The weighted three-month rate is less sensitive to single-month variation. Year-over-year percentage-point changes are included where a 12-month comparison exists.

Gaps mean absent market-months; changing coverage limits global comparisons.



The final order month contains 2,037 eligible orders, from 2,123 total distinct orders. Its maximum order date is 2018-01-31; shipments continue into 2018-02-06. Order date and outcomes are present for the final month, but data extraction completeness cannot be inferred from non-missingness alone. Excluding that month, the latest weighted rolling rate is 57.3% in Dec 2017.

Across the full period, the rate is 57.3%; excluding January 2018 it is 57.3%. Quarterly rates range from 56.5% to 58.7%; the final 2018Q1 value reflects January only. The final-month Order Status mix is reported in the CSV and workbook, but labels other than COMPLETE/CLOSED are not automatically interpreted as unresolved.

5. Reliability over time - robustness

Geographic coverage changes over time, limiting global comparisons. January 2018 contains only Pacific Asia; absent market-months are not zero late rates.

Changing market coverage; 2018Q1 contains January / Pacific Asia only.

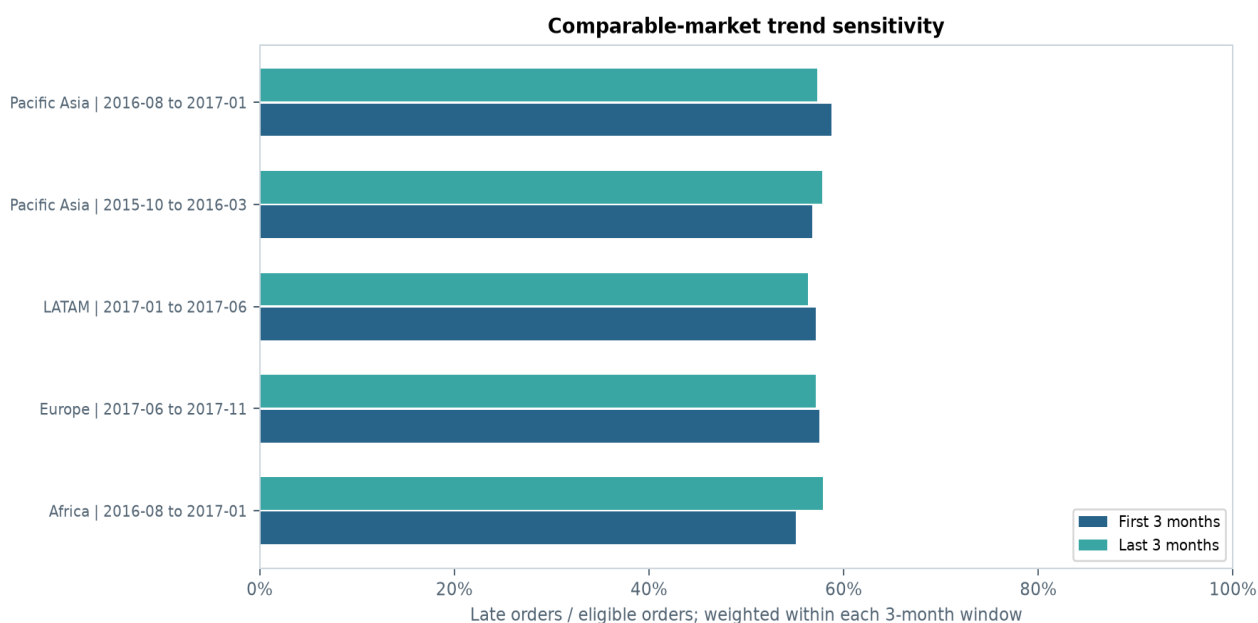


The full-year rates are 57.2%, 57.6%, and 57.0% in 2015-2017. Applying common mode weights gives 57.2%, 57.5%, and 57.2%, which does not demonstrate a sustained trend. January 2018 is 58.7%, up 0.96 percentage points month over month and 2.88 points year over year, but its Wilson interval overlaps nearby months. January has 2,123 total orders, including 86 canceled; its other-than-COMPLETE/CLOSED share is 55.0%, versus 55.8% in December. These labels alone do not prove unresolved shipments. Monthly and quarterly patterns and final-month exclusion support cautious monitoring rather than a deterioration claim. January 2018 contains only Pacific Asia, so its global rate is a Pacific Asia rate rather than a geographically comparable global cohort. Market coverage changes substantially across the extract; missing market-months must not be treated as zero lateness.

What this means in simple terms... Reliability is not a single flat number over time. Monthly rates show short-term movement, while rolling and quarterly rates give a steadier view; the ending month deserves a completeness check.

5. Comparable-market sensitivity

Within each market, include every maximal continuous run of at least six calendar months with at least 100 eligible orders in every month. Compare count-weighted first-three-month and last-three-month late rates. No market covers the full extract consistently; mode and region mix can still change.



Same market; >=6 continuous months; >=100 eligible orders/month. Mix can still change.

Market / window	First 3m late / eligible	Last 3m late / eligible	Change (pp)
Africa 2016-08 to 2017-01	961/1,745 (55.1%)	1,125/1,944 (57.9%)	+2.80
Europe 2017-06 to 2017-11	2,350/4,087 (57.5%)	2,458/4,303 (57.1%)	-0.38
LATAM 2017-01 to 2017-06	2,266/3,967 (57.1%)	2,390/4,244 (56.3%)	-0.81
Pacific Asia 2015-10 to 2016-03	2,174/3,827 (56.8%)	2,878/4,977 (57.8%)	+1.02
Pacific Asia 2016-08 to 2017-01	723/1,231 (58.7%)	797/1,391 (57.3%)	-1.44

All qualifying windows are included; the rule uses order counts and calendar continuity, not late-rate outcomes. No market has qualifying continuous coverage throughout January 2015-January 2018. These shorter within-market comparisons do not rescue the full-period global comparison; region, shipping-mode and order mix can still change.

What this means in simple terms... Compare the same market over well-covered windows before interpreting changes. The pooled global line combines changing markets and cannot establish a like-for-like global trend.

6. Combined operational insights & recommendations

The results support targeted investigation rather than a blanket operating change. Highest-rate geographies, high-volume late contributors, mode-specific gaps, and trend changes are all measurable; the underlying reasons are not in this dataset.

1. Prioritize investigation by late-order volume. Begin with Europe and Western Europe because their late-order counts are largest. These are priorities because of late-order volume, not evidence of unusually high rates. Validate with local shipment and delivery-arrival records before assigning causes. Demonstrated: group-level rate/count differences. Follow-up hypothesis: operational drivers vary by lane.
2. Validate First Class and Same Day measurement conventions. Reconcile scheduled days, actual-day rounding, delivery status, and shipment/delivery timestamps; confirm Same Day rounding and distinguish shipment from arrival events. Demonstrated: 12-hour shipment timestamp intervals coexist with actual-day values of 0/1 and complete agreement between late labels and positive gaps in eligible orders. Follow-up hypothesis: definition or event-timing differences contribute to the mismatch.
3. Monitor reliability with an order-month control view. Use the weighted three-month rate alongside rates and counts by market and mode; validate end-of-period extract completeness before escalating a trend. Demonstrated: monthly/quarterly movement and changing mode shares. Follow-up hypothesis: changes reflect service/process shifts rather than data capture or mix.

These recommendations require operational follow-up; they are not proven causal levers. Sales on late orders are not lost revenue; suspected-fraud status is not evidence of confirmed fraud.

7. Supporting tables & methodological note

Mode	Eligible	Late	Late rate	Median gap (days)	Late median gap
First Class	9,602	9,602	100.0%	1.00	1.00
Second Class	12,256	9,803	80.0%	2.00	3.00
Same Day	3,407	1,648	48.4%	0.00	1.00
Standard Class	37,632	14,995	39.8%	0.00	2.00

COMPLETE/CLOSED sensitivity: 28,965 eligible orders, late rate 57.4% (primary baseline 57.3%); 28,965 of 65,752 total orders are COMPLETE/CLOSED. The sensitivity does not replace primary results because the remaining order statuses are not necessarily unresolved or invalid shipping outcomes.

Quarterly check: 2015Q1 is 57.2%; 2018Q1 is 58.7%. Comparisons use the same order-level denominator and label. The monthly and quarterly views should be reviewed together rather than selecting whichever appears more favorable.

Project role: the project owner guided the business questions and reviewed the findings. AI-assisted Python implementation generated the analysis and dashboard.

Reproducibility: run ``python analyze.py --portfolio`` from the project folder. It validates source assumptions and regenerates CSV tables, charts, workbook and the portfolio report. ``streamlit run dashboard.py`` launches the dashboard locally. Source data are unchanged.

Dataset citation: Constante, Fabian; Silva, Fernando; Pereira, António. DataCo Smart Supply Chain for Big Data Analysis, Mendeley Data, v5, 12 March 2019, <https://data.mendeley.com/datasets/8gx2fvg2k6/5>, CC BY 4.0.

Appendix — supporting visuals

The following charts summarize the changing share of each shipping mode among eligible distinct orders and the geographic spread of late rates. See the monthly and quarterly CSV tables for the counts and rates behind the graphics.

Changing market coverage also affects mode mix; this is not a fixed global cohort.

